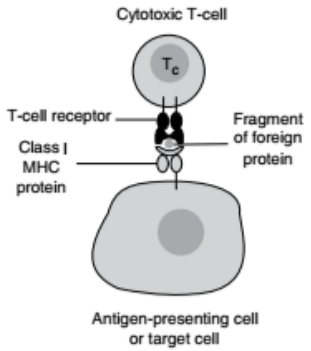
McKeithan's kinetic proofreading model provides mechanism for ligand discrimination



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Immunology

Kinetic proofreading in T-cell receptor signal transduction

(protein-tyrosine kinase/major histocompatibility complex/mathematical model)

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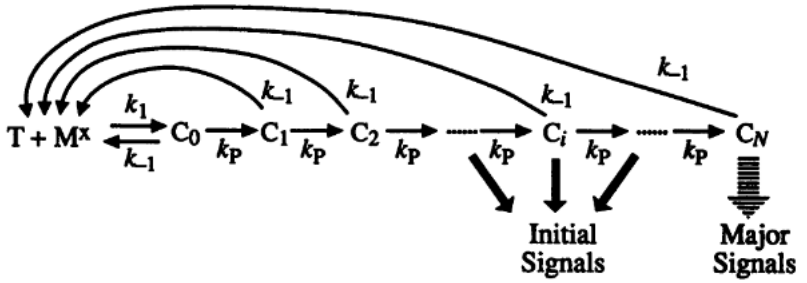


FIG. 1. Proofreading scheme. Nascent complexes C_0, formed from the TCR complex (T) and peptide x–MHC (M^x), must undergo N modifications, each with rate constant k_P, before generating the active complex C_N. At every step, the complex may dissociate with rate k_{-1}, leading to complete reversion of the subunits to their unmodified forms. For full activation, signals must be generated from the final complex C_N. Other signals may be generated from earlier complexes.

Key idea: delay allows for error correction

Example: Delay of 3 sec can amplify difference between ligands with 10-fold difference in off-rate by approximately 1 million-fold.
Error rate is given by

$$F = e^{-(\Delta k_{\mathrm{off}} \tau)}$$